Crop Classification Using Satellite Data

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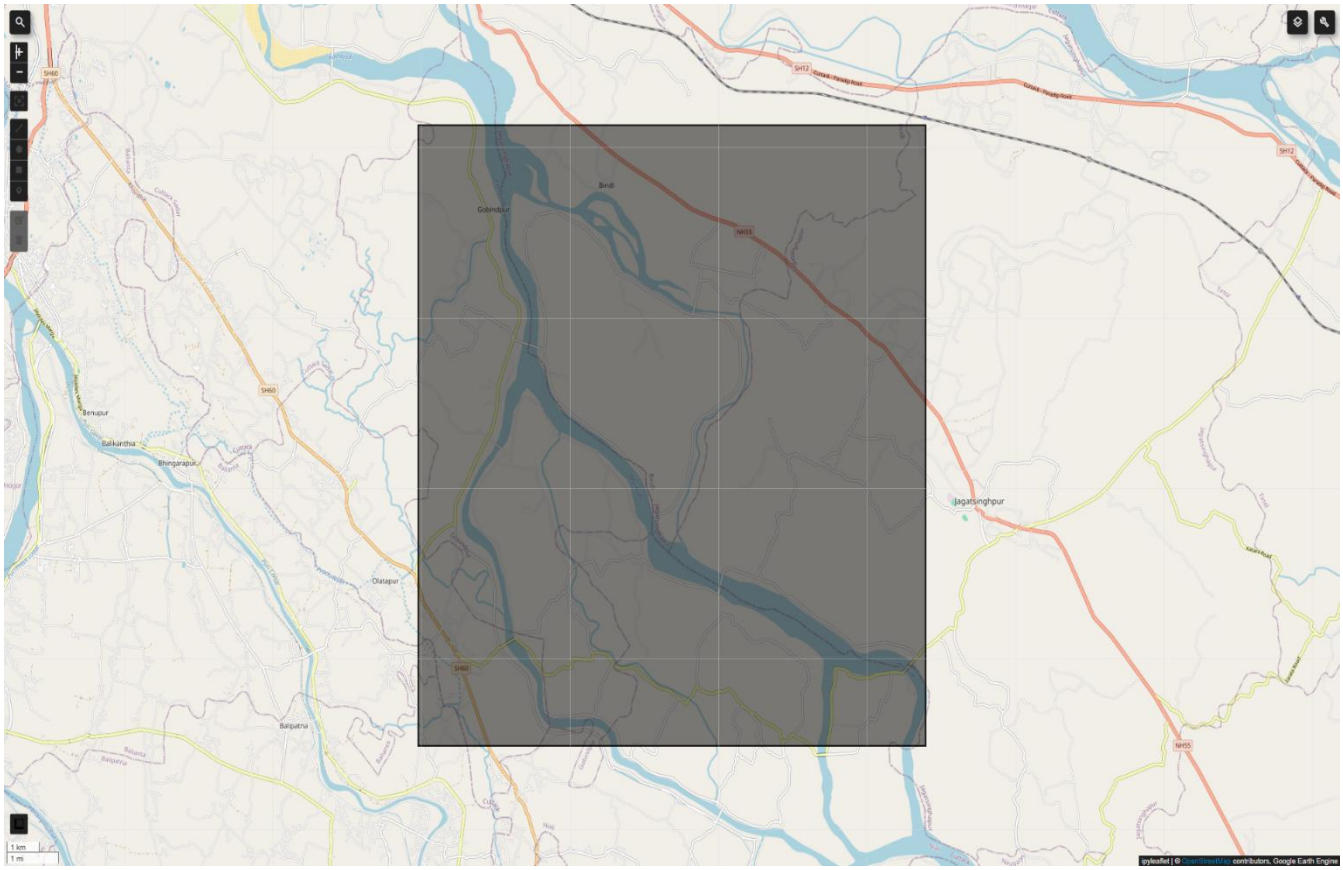
Tools Used: Google Earth Engine, Python, Random Forest Classifier

1. Objective

Objective The main goal of this project was to create a strong, cloud-based machine learning pipeline to automate the classification of land-cover types. By using Sentinel-2 multi-spectral data and the Google Earth Engine platform, the model was designed to clearly distinguish between river features, sparse vegetation, and dense agricultural crops in a complex delta landscape.

2. Study Area and AOI Definition

Study Area and AOI Definition The research focused on the **Mahanadi River Delta** in the Jagatsinghpur District of Odisha. This location was chosen because of its varied geography, which includes active river channels, different levels of agricultural activity, and thick perennial vegetation.



3. Data Processing and Feature Engineering

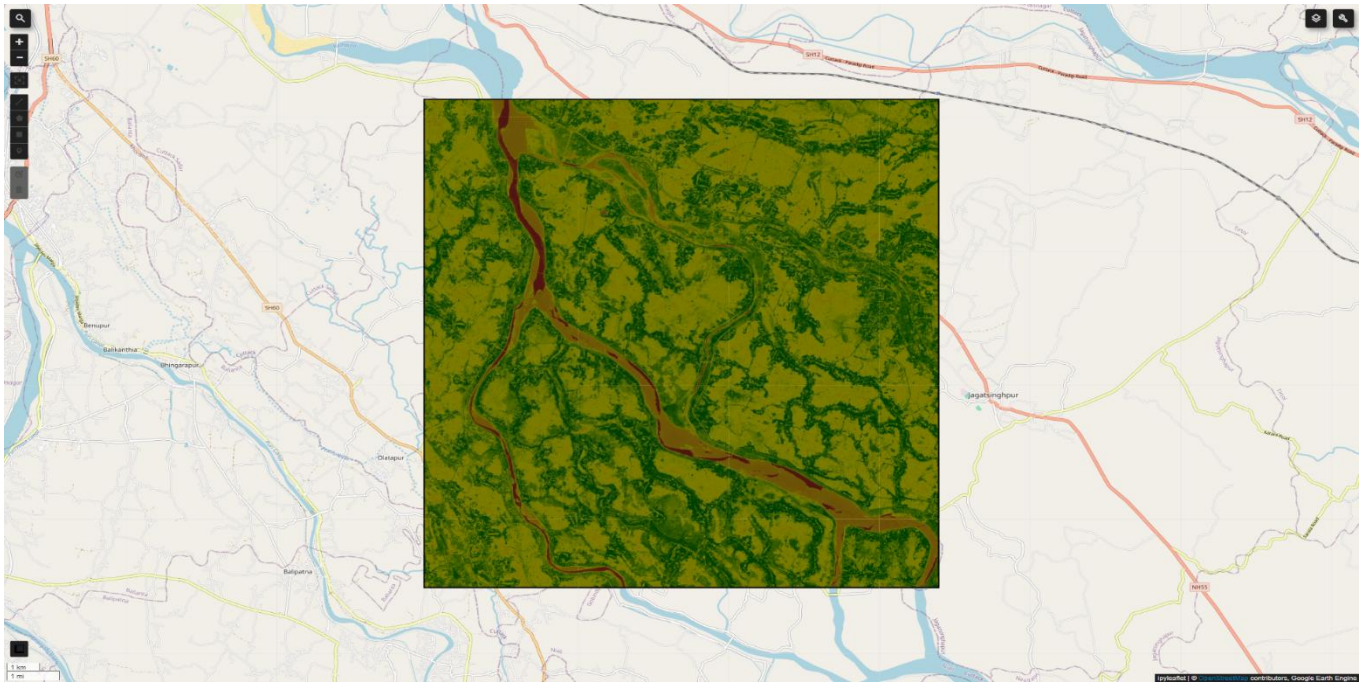
Satellite Source: Sentinel-2 Surface Reflectance (Harmonized) data with a 10-meter spatial resolution.

Time Period: A median composite created from November 2024 to February 2025 to capture peak agricultural health.

Spectral Features: Six spectral bands (B2, B3, B4, B8, B11, B12) and a calculated **NDVI.**

Vegetation Index Calculation:

$$NDVI = \frac{B8(NIR) - B4(Red)}{B8(NIR) + B4(Red)}$$



4. Training and Classification

Manual ground-truth data was collected using precise polygons across three distinct classes directly on the Sentinel-2 RGB composite.

- **Class 0: Non-vegetation (Sand)** – Focused on the dry riverbed.
- **Class 1: Sparse Vegetation** – Focused on early-stage or harvested crops.
- **Class 2: Dense Vegetation** – Focused on high-density canopies and healthy crops.

The dataset was divided into **70% for training**, which includes **4,739 samples**, and **30% for validation**, which includes **2,017 samples**.

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Training samples: 4739
Testing samples: 2017
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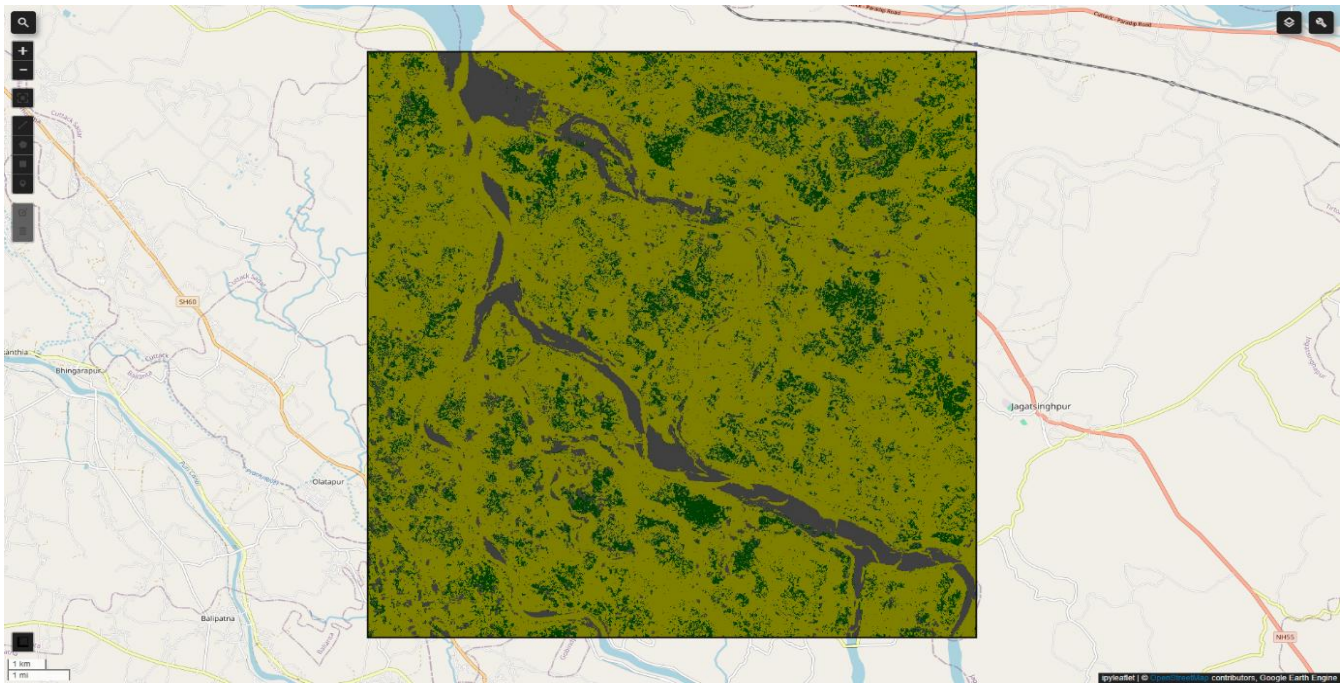
5. Results and Performance

The Random Forest model, which has 50 trees, successfully applied the training logic across the entire study area.



Quantitative Evaluation

The model achieved an Overall Accuracy of 94.15% and a Kappa Score of 0.8967.

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=== CLASSIFICATION METRICS ===
Overall Accuracy : 94.15%
Kappa Score      : 0.8967

=== CONFUSION MATRIX ===
                        Predicted Non-veg  Predicted Sparse  Predicted Dense
Non-vegetation (Sand)                 147                 0                1
Sparse Vegetation                       2               893               52
Dense Vegetation                        6                57              859
```

6. Discussion and Conclusion

The high accuracy shows that Random Forest is a useful tool for regional crop monitoring. The small misclassifications observed in the confusion matrix between sparse and dense vegetation are probably due to the **10-meter pixel resolution**. Here, "mixed pixels" at the edges of fields have characteristics of both classes.

The final outputs were saved to Google Drive for further analysis.

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Export started. Confirm the tasks in the GEE Code Editor to finalize.
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